

CELEBAL EXCELLENCE INTERNSHIP 2026

Week 4 Assignment

Azure Cloud Fundamentals and
Data Pipeline Implementation using ADF

Submitted By: Shubham Sharma

Dataset: Sample - Superstore.csv

Date: 15 June 2026

ADF Instance: ADF-Shubham16

Objective

To understand Azure cloud concepts and build a complete end-to-end data pipeline using Azure Storage Account and Azure Data Factory (ADF). The assignment covers resource provisioning, pipeline design, metadata validation, IAM role assignment, and pipeline execution monitoring.

Task 1 — Azure Portal & Resource Group

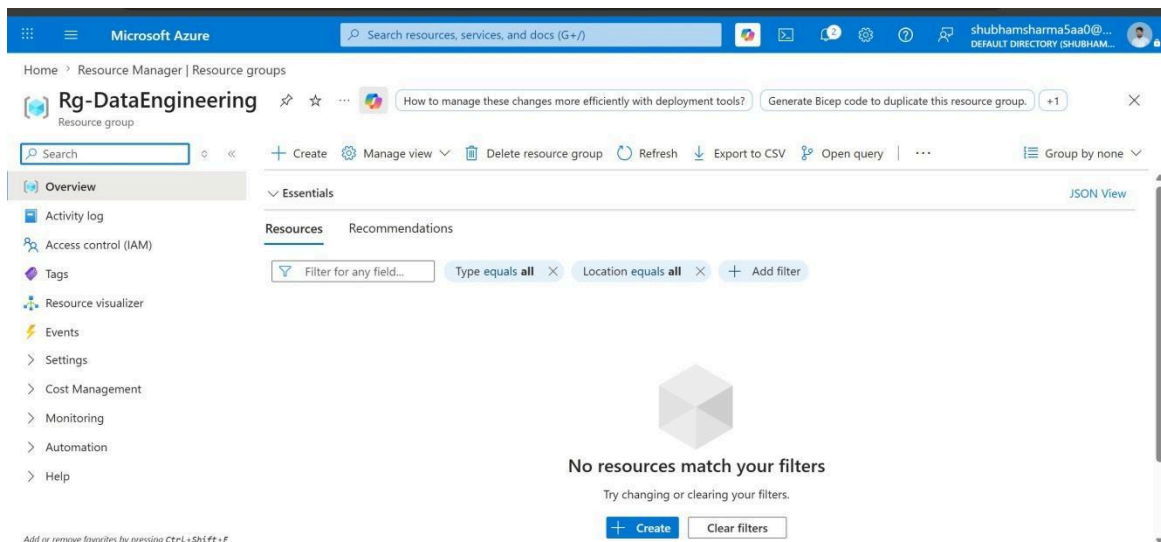
Steps Performed

1. Logged into Azure Portal at portal.azure.com
2. Navigated to Resource Groups via the search bar
3. Clicked + Create and filled in the configuration below
4. Clicked Review + Create → Create
5. Verified Resource Group appeared in the list after deployment

Field	Value
Subscription	Azure for Students
Resource Group Name	Rg-DataEngineering
Region	Central India

Deliverable

Screenshot — Resource Group — Rg-DataEngineering created successfully



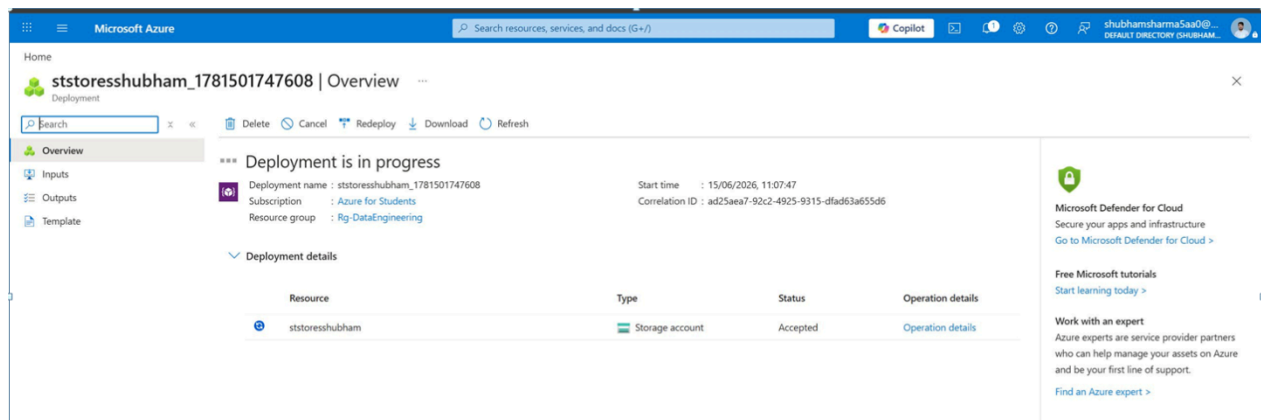
Task 2 — Storage Account Setup

2a. Create Storage Account

1. Searched for Storage Accounts in Azure Portal → clicked + Create
2. Filled in the configuration below and clicked Review + Create → Create

Field	Value
Resource Group	Rg-DataEngineering
Storage Account Name	ststoresshubham
Region	Central India
Performance	Standard
Redundancy	LRS (Locally Redundant Storage)

Screenshot — Storage Account ststoresshubham — Deployment



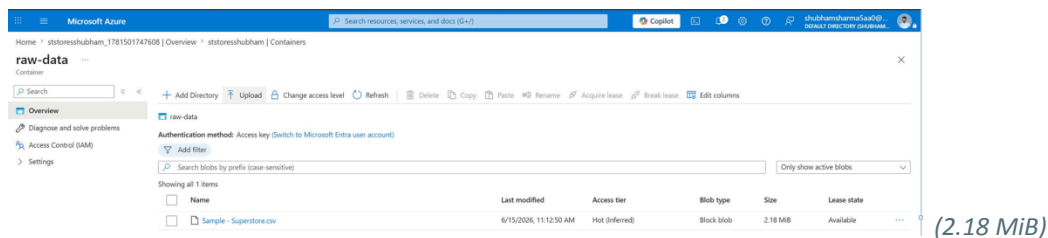
2b. Create Blob Container

1. Inside the Storage Account → navigated to Containers → + Container
2. Created container: raw-data (for source CSV and processed output)
3. Access level kept as Private (no anonymous access)

2c. Upload CSV File

1. Opened raw-data container → clicked Upload
2. Selected and uploaded: Sample - Superstore.csv
3. Verified file appeared with correct size (2.18 MiB) and last-modified timestamp

Screenshot — Blob Container raw-data with Sample - Superstore.csv uploaded



Task 3 — Azure Data Factory Setup

3a. Create Azure Data Factory

1. Searched for Data Factories in Azure Portal → clicked + Create

Field	Value
Resource Group	Rg-DataEngineering
ADF Name	ADF-Shubham16
Region	Central India
Version	V2

2. Clicked Review + Create → Create (deployment took ~2–3 minutes)
3. After deployment, clicked Launch Studio to open ADF UI

3b. ADF UI Exploration

The ADF Studio consists of three main sections:

Tab	Purpose
Author	Create/edit pipelines, datasets, linked services
Monitor	View pipeline execution status, debug logs, trigger runs
Manage	Manage linked services, integration runtimes, Git config

3c. Create Linked Service

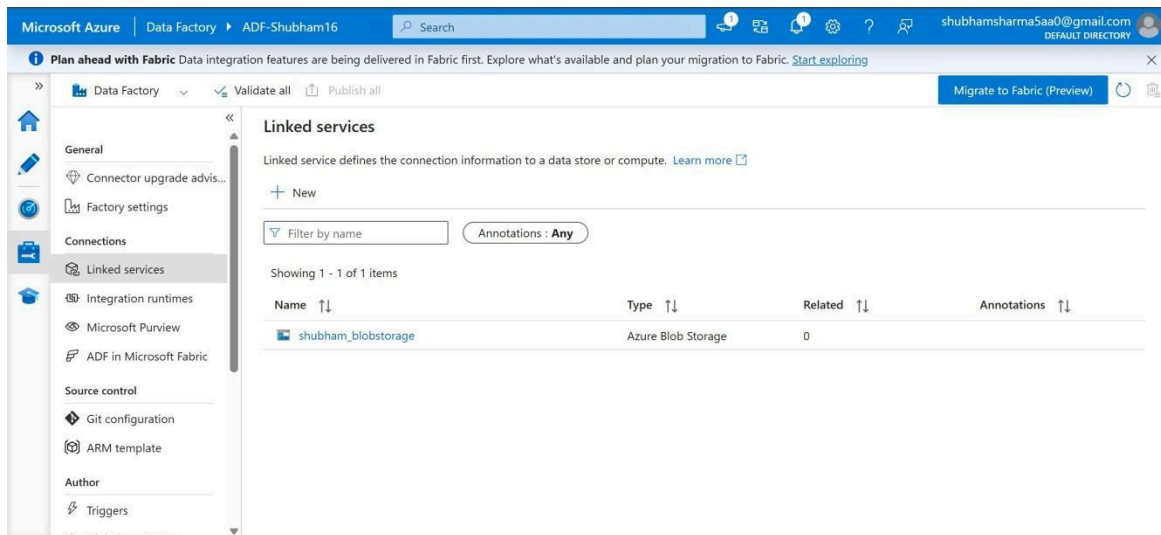
1. Navigated to Manage tab → Linked Services → + New
2. Selected Azure Blob Storage → Continue

Field	Value
Linked Service Name	shubham_blobstorage
Storage Account	ststoreshubham
Authentication	Account Key (auto-populated)

1. Clicked Test Connection → received Connection Successful confirmation

- Clicked Create to save

Screenshot — Linked Service — shubham_blobstorage (Azure Blob Storage)

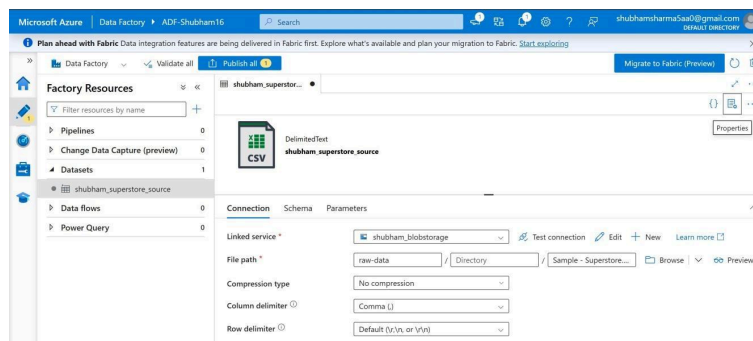


3d. Create Datasets

Source Dataset

Field	Value
Dataset Name	shubham_superstore_source
Linked Service	shubham_blobstorage
Container	raw-data
File	Sample - Superstore.csv
First Row as Header	Yes

Screenshot — Source Dataset — shubham_superstore_source

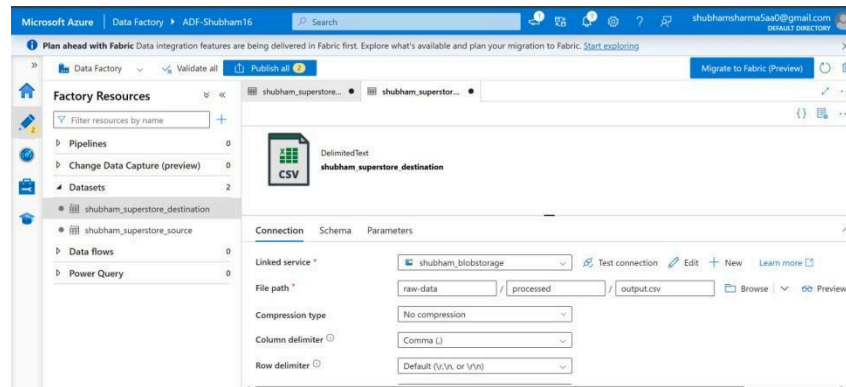


Destination Dataset

Field	Value
Dataset Name	shubham_superstore_destination

Linked Service	shubham_blobstorage
Container	raw-data
Folder	processed
File	output.csv
First Row as Header	Yes

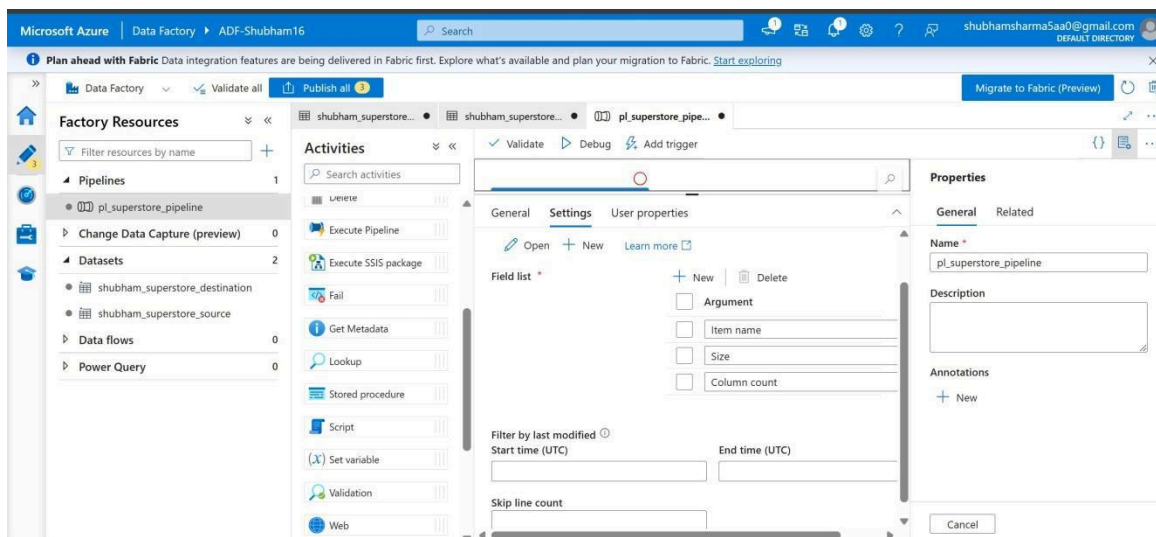
Screenshot — Destination Dataset — shubham_superstore_destination



3e. Get Metadata Activity

1. Created a new pipeline: pl_superstore_pipeline
2. Dragged Get Metadata activity from the Activities panel onto the canvas
3. In Settings tab, selected dataset: shubham_superstore_source
4. Added the following field list items:
 - itemName — retrieves file name
 - size — retrieves file size in bytes
 - columnCount — retrieves number of columns in the file

Screenshot — Get Metadata Activity — Field list: itemName, size, columnCount



Task 4 — Pipeline Development with Copy Data Activity

Steps Performed

1. In the same pipeline pl_superstore_pipeline, dragged Copy Data activity onto the canvas
2. Connected Get Metadata → Copy Data using the green success arrow
3. This ensures metadata is validated before copying begins

Copy Data Configuration

Source Tab

Dataset	shubham_superstore_source
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Sink Tab

Field	Value
Dataset	shubham_superstore_destination
Copy Behavior	PreserveHierarchy

Mapping Tab

1. Clicked Import Schemas — all 21 columns from Superstore CSV auto-mapped
2. Verified column mapping: Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country, City, State, Postal Code, Region, Product ID, Category, Sub-Category, Product Name, Sales, Quantity, Discount, Profit

Pipeline Architecture

Screenshot — Pipeline Design — Get Metadata → Copy Data (both activities connected)

Task 5 — Pipeline Execution

Debug Run

1. Clicked Debug button in the pipeline toolbar to run without publishing
2. Monitored real-time execution in the Output tab at the bottom

Activity	Status	Result
Get Metadata	✓ Succeeded	Returned itemName, size, columnCount
Copy Data	✓ Succeeded	Rows read & written: 9994 Duration: 15s

Screenshot — Pipeline Execution — Both activities Succeeded (Get Metadata: 12s, Copy Data: 15s)

The screenshot shows the Microsoft Azure Data Factory console. The pipeline 'pl_superstore_pipeline' is in a 'Succeeded' state. The 'Output' tab displays the following table:

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Copy data1	✓ Succeeded	Copy data	6/15/2026, 11:51:20 AM	15s	AutoResolveIntegrationRuntime (Central India)
Get Metadata1	✓ Succeeded	Get Metadata	6/15/2026, 11:51:08 AM	12s	AutoResolveIntegrationRuntime (Central India)

Output Verification

1. Navigated to Storage Account → raw-data container → processed folder
2. Confirmed file output.csv was created successfully
3. Verified file size matched the source CSV

Task 6 — IAM Role Assignment

Steps Performed

1. Navigated to Storage Account (ststoresshubham) → Access Control (IAM)
2. Clicked + Add → Add role assignment for each of the following roles:

Role Assignments

Role	Assigned To	Purpose
Reader	Shubham Sharma (User)	Read-only access to storage resources
Contributor	Shubham Sharma (User)	Create and manage storage resources
Storage Blob Data Contributor	ADF-Shubham16 (Managed Identity)	ADF can read/write to Blob Storage

Screenshot — IAM Role Assignments — Reader, Contributor, Storage Blob Data Contributor (ADF Managed Identity)

The screenshot displays the Microsoft Azure portal interface for the 'ststoresshubham' storage account. The 'Access Control (IAM)' section is active, showing a list of role assignments. The interface includes a search bar, filters for Type, Role, Scope, and Group by, and a table of assignments. The table lists four roles: Owner (Shubham Sharma), Contributor (Shubham Sharma), Reader (Shubham Sharma), and Storage Blob Data Contributor (ADF-Shubham16 Managed Identity). The left sidebar shows the navigation menu with 'Access Control (IAM)' selected. The top bar shows the user profile and search bar.

Name	Type	Role	Scope	Condition
shubham sharma e60f4a24-16f3-46d2-8580-6bc986f...	User	Owner	Subscription (Inherited)	None
shubham sharma e60f4a24-16f3-46d2-8580-6bc986f...	User	Contributor	This resource	None
shubham sharma e60f4a24-16f3-46d2-8580-6bc986f...	User	Reader	This resource	None
ADF-Shubham16 5e293304-e4b8-46e7-b12c-bf64f...	Managed identity	Storage Blob Data Contributor	This resource	Add

Mini Project — End-to-End Pipeline

Problem Statement

Build a complete pipeline that reads a CSV file from Azure Blob Storage and processes it using Azure Data Factory, copying data from source to destination with metadata validation.

Architecture Overview

CSV File (Blob Storage: raw-data) → Linked Service (shubham_blobstorage) → Source Dataset → [Get Metadata → Copy Data] → Destination Dataset → output.csv (raw-data/processed/)

Requirements Fulfilled

Requirement	Implementation	Status
Source: CSV in Blob	Sample - Superstore.csv in raw-data container	✓
Linked Service	shubham_blobstorage (Azure Blob Storage)	✓
Source Dataset	shubham_superstore_source (DelimitedText, 21 columns)	✓
Destination Dataset	shubham_superstore_destination (raw-data/processed)	✓
Metadata Check	Get Metadata — itemName, size, columnCount	✓
Copy Data Activity	Full 21-column copy with schema mapping	✓
Pipeline Executed	Debug run — both activities Succeeded	✓
Data at Destination	output.csv created in raw-data/processed/	✓

Screenshot — Final Pipeline Execution — Succeeded

The screenshot displays the Microsoft Azure Data Factory console. The left sidebar shows the 'Factory Resources' tree with 'Pipelines' expanded, listing 'pl_superstore_pipeline'. The main workspace shows the pipeline diagram with two activities: 'Get Metadata1' and 'Copy data1'. Below the diagram, the 'Pipeline run' summary indicates a successful execution with the ID '5086ced0-2bae-423c-9e2c-772ea7ebd53a'. A table lists the activities and their status:

Activity name	Activity ID	Activity status	Activity name	Run start	Duration	Integration runtime
Copy data1	5086ced0-2bae-423c-9e2c-772ea7ebd53a	Succeeded	Copy data	6/15/2026, 11:51:20 AM	15s	AutoResolveIntegrationRuntime (Central India)
Get Metadata1	5086ced0-2bae-423c-9e2c-772ea7ebd53a	Succeeded	Get Metadata	6/15/2026, 11:51:08 AM	12s	AutoResolveIntegrationRuntime (Central India)

The right sidebar shows the 'Properties' panel for the pipeline, with the 'Name' field set to 'pl_superstore_pipeline'.

Summary

This assignment provided hands-on experience with Azure cloud services and the Azure Data Factory pipeline ecosystem. Key learnings:

- Set up the Azure environment from scratch by creating a Resource Group (Rg-DataEngineering), Storage Account (ststoresshubham), and Blob Container (raw-data) through the Azure Portal.
- Explored the Azure Data Factory Studio and its three core sections: Author (pipeline design), Monitor (run tracking), and Manage (linked services and configuration).
- Configured a Linked Service (shubham_blobstorage) to establish a secure connection between ADF and the Azure Blob Storage account using Account Key authentication.
- Defined two datasets in DelimitedText (CSV) format: shubham_superstore_source pointing to the uploaded Superstore file, and shubham_superstore_destination for the pipeline output.
- Added a Get Metadata activity to the pipeline that retrieves key file attributes (itemName, size, columnCount) from the source CSV before any data movement begins.
- Built the pipeline pl_superstore_pipeline by chaining Get Metadata and Copy Data activities with a success dependency arrow, ensuring metadata is checked before data is moved.
- Configured the Copy Data activity with source and sink datasets and imported the full 21-column schema from the Superstore CSV.
- Ran the pipeline using the Debug trigger and monitored the execution in the Output tab, with both Get Metadata (12s) and Copy Data (15s) activities completing successfully.
- Assigned Reader and Contributor roles to the user account via IAM, and granted the Storage Blob Data Contributor role to the ADF Managed Identity (ADF-Shubham16) for secure access to Blob Storage.
- Verified the end-to-end pipeline output by checking the raw-data/processed/ folder, where output.csv was successfully created with all 9,994 rows copied from the source.